CSA5108 Cryptography and Network security

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Exp 9 : Playfair Cipher — PT-109 Message Decryption
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char matrix[5][5] = {
   {'M','O','N','A','R'},
   {'C','H','Y','B','D'},
   {'E','F','G','I','K'},
   {'L','P','Q','S','T'},
   {'U','V','W','X','Z'}
};

void findPosition(char c, int *r, int *col) {
   if (c == 'J') c = 'I';

   for (int i = 0; i < 5; i++)
      for (int j = 0; j < 5; j++)
         if (matrix[i][j] == c) {
            *r = i;
            *col = j;
            return;
         }
}

int main(void) {
   char cipher[500];
   int n = 0;

   printf("Enter ciphertext: ");
   fgets(cipher, sizeof(cipher), stdin);
```

```
    for (int i = 0; cipher[i]; i++)
        if (isalpha((unsigned char)cipher[i]))
            cipher[n++] = toupper((unsigned char)cipher[i]);

    cipher[n] = '\0';

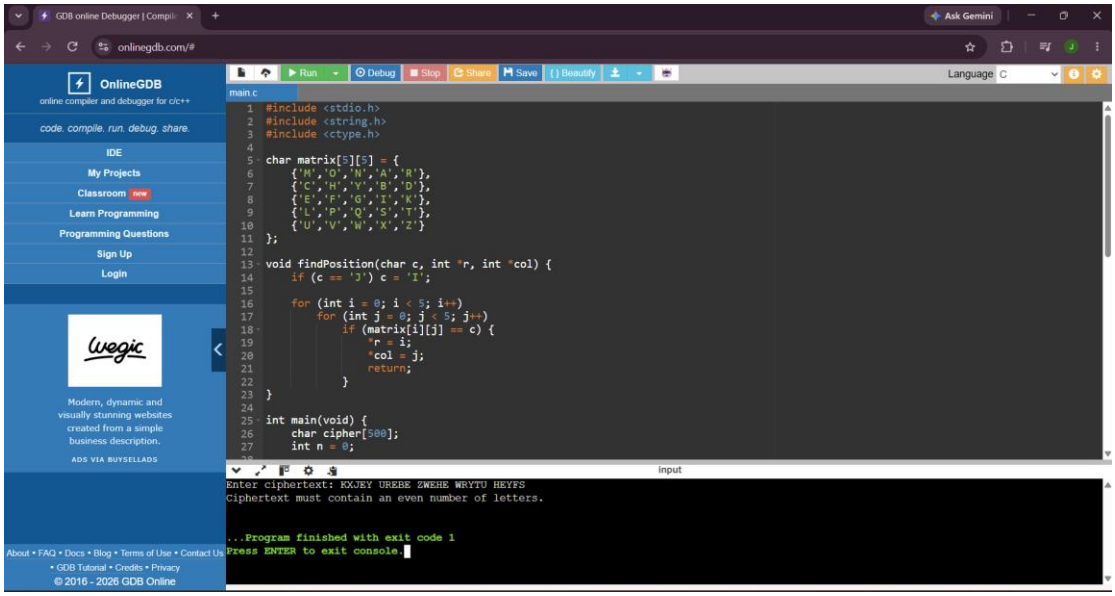
    if (n % 2 != 0) {
        printf("Ciphertext must contain an even number of letters.\n");
        return 1;
    }

    for (int i = 0; i < n; i += 2) {
        int r1, c1, r2, c2;
        char a = cipher[i], b = cipher[i + 1];

        findPosition(a, &r1, &c1);
        findPosition(b, &r2, &c2);

        if (r1 == r2) {
            cipher[i] = matrix[r1][(c1 + 4) % 5];
            cipher[i + 1] = matrix[r2][(c2 + 4) % 5];
        } else if (c1 == c2) {
            cipher[i] = matrix[(r1 + 4) % 5][c1];
            cipher[i + 1] = matrix[(r2 + 4) % 5][c2];
        } else {
            cipher[i] = matrix[r1][c2];
            cipher[i + 1] = matrix[r2][c1];
        }
    }

    printf("Decrypted text: %s\n", cipher);
    return 0;
}
```


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Language C
main.c
#include <stdio.h>
#include <string.h>
#include <ctype.h>
char matrix[5][5] = {
{'M','O','N','A','R'},
{'C','H','Y','B','D'},
{'E','F','G','I','K'},
{'L','P','Q','S','T'},
{'U','V','W','X','Z'}
};
void findPosition(char c, int *r, int *col) {
if (c == 'J') c = 'I';
for (int i = 0; i < 5; i++)
for (int j = 0; j < 5; j++)
if (matrix[i][j] == c) {
*r = i;
*col = j;
return;
}
}
int main(void) {
char cipher[500];
int n = 0;
input
Enter ciphertext: KXJEY UREBE ZWEHE WRYTU HEYFS
Ciphertext must contain an even number of letters.
...Program finished with exit code 1
Press ENTER to exit console.